

# Midterm Project

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## **1 Question 1:**

### **1.1 Flat Random Sounds**

The music generated with the requisite parameters (by default in intervals of 40 sounds in three voices each, at least “16 measures” long) is displayed below:

Musical score for a three-staff instrument in 3/8 time. The score is divided into four systems, each containing three staves. The key signature has one sharp (F#). The piece includes various dynamics and fingerings.

**System 1 (Measures 1-11):**

- Staff 1: Measures 1-11. Dynamics: *f*, *ff*. Fingerings: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11.
- Staff 2: Measures 1-11. Dynamics: *p*. Fingerings: 1, 2, 3, 4, 5, 6, 7, 8, 9.
- Staff 3: Measures 1-11. Dynamics: *f*, *mf*. Fingerings: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12.

**System 2 (Measures 12-21):**

- Staff 1: Measures 12-21. Dynamics: *pp*. Fingerings: 12, 13, 14, 15, 16, 17, 18.
- Staff 2: Measures 12-21. Dynamics: *pp*, *p*. Fingerings: 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21.
- Staff 3: Measures 12-21. Dynamics: *f*. Fingerings: 13, 14, 15, 16, 17, 18, 19.

**System 3 (Measures 22-31):**

- Staff 1: Measures 22-31. Dynamics: *mp*, *pp*, *mf*, *mp*, *mp*. Fingerings: 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37.
- Staff 2: Measures 22-31. Dynamics: *mp*, *f*. Fingerings: 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35.
- Staff 3: Measures 22-31. Dynamics: *mf*, *mp*, *f*. Fingerings: 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31.

**System 4 (Measures 32-40):**

- Staff 1: Measures 32-40. Dynamics: *ff*, *ff*. Fingerings: 38, 39, 40.
- Staff 2: Measures 32-40. Dynamics: *ff*. Fingerings: 36, 37, 38, 39, 40.
- Staff 3: Measures 32-40. Dynamics: *mf*. Fingerings: 32, 33, 34, 35, 36, 37, 38, 39, 40.

## 2 Question 2:

### 2.1 Markov Bank

The bank of pitches (default the first 12 sounds) used in markov chain generation is:

A musical score for three staves, each in treble clef with a key signature of one sharp (F#) and a common time signature (C). The score consists of 12 measures, each containing a single pitch. The pitches are numbered 1 through 12. The first staff has dynamic markings *fff* at measure 4 and *pp* at measure 10. The second staff has dynamic markings *p* at measure 2 and *pp* at measure 10. The third staff has dynamic markings *f* at measure 4, *mf* at measure 8, and *p* at measure 10.

### 2.2 Markov Chain Table

The table generated for the Markov Chain is:

| x  | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   | 11   | 12   |
|----|------|------|------|------|------|------|------|------|------|------|------|------|
| 1  | 0.1  | 0.06 | 0.09 | 0.06 | 0.0  | 0.12 | 0.11 | 0.07 | 0.07 | 0.12 | 0.05 | 0.15 |
| 2  | 0.06 | 0.09 | 0.14 | 0.16 | 0.01 | 0.0  | 0.04 | 0.05 | 0.02 | 0.16 | 0.05 | 0.22 |
| 3  | 0.05 | 0.04 | 0.09 | 0.12 | 0.07 | 0.04 | 0.12 | 0.03 | 0.1  | 0.12 | 0.06 | 0.16 |
| 4  | 0.06 | 0.01 | 0.17 | 0.14 | 0.1  | 0.0  | 0.15 | 0.12 | 0.04 | 0.02 | 0.11 | 0.08 |
| 5  | 0.13 | 0.04 | 0.18 | 0.01 | 0.12 | 0.03 | 0.04 | 0.05 | 0.15 | 0.15 | 0.0  | 0.1  |
| 6  | 0.11 | 0.1  | 0.02 | 0.1  | 0.11 | 0.1  | 0.07 | 0.07 | 0.13 | 0.01 | 0.1  | 0.08 |
| 7  | 0.03 | 0.11 | 0.11 | 0.1  | 0.09 | 0.09 | 0.1  | 0.07 | 0.03 | 0.09 | 0.0  | 0.18 |
| 8  | 0.07 | 0.04 | 0.01 | 0.13 | 0.03 | 0.08 | 0.14 | 0.1  | 0.08 | 0.08 | 0.08 | 0.16 |
| 9  | 0.01 | 0.12 | 0.06 | 0.02 | 0.2  | 0.07 | 0.04 | 0.11 | 0.04 | 0.07 | 0.16 | 0.1  |
| 10 | 0.04 | 0.0  | 0.13 | 0.1  | 0.09 | 0.02 | 0.08 | 0.04 | 0.13 | 0.11 | 0.13 | 0.13 |
| 11 | 0.1  | 0.11 | 0.04 | 0.09 | 0.05 | 0.07 | 0.1  | 0.09 | 0.11 | 0.06 | 0.03 | 0.15 |
| 12 | 0.0  | 0.15 | 0.06 | 0.15 | 0.04 | 0.05 | 0.11 | 0.1  | 0.13 | 0.04 | 0.07 | 0.1  |

## 2.3 Markov Chain

The music generated with the requisite parameters (by default in intervals of 20 sounds, at least "16 measures" long) by Markov Chain is displayed below:



### 3 Methodology

The music is generated by the `midterm.py` script in the `./python` folder. The length of the **Flat Random Sounds**, the number of voices, the length of the **Markov Bank**, and the length of the **Markov Chain** are all configurable.

To generate a new `midterm.pdf` document, run the script `./generate.sh` .  
The usage for this command is detailed by running `./generate.sh -h`  
or `./generate.sh --help`

The generated document is available as `assignment-print.pdf` in the root directory.